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Memory Moves Markets

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1 Big picture

- **Question.** Can memory associations inside investors' minds create attention, trading pressure, and price movements in financial markets?
- **Main idea.** Attention is usually studied as externally triggered by news, media, abnormal returns, volume, or salient displays. This paper studies internally generated attention: one firm cues investors to recall another firm associated with it in memory.
- **Empirical innovation.** The paper identifies memory links using rigid earnings-announcement schedules of Pattern firms and exogenous calendar rotations. If two firms announced earnings on the same day in one quarter because of calendar mechanics, then one firm's later announcement can cue the other firm.
- **Core treatment.** A firm receives a memory cue on day t when a firm that announced earnings on the same day in the previous fiscal quarter announces earnings on day t .
- **Main price result.** Cued firms earn positive characteristic-adjusted abnormal returns. The paper reports effects around 4–6 basis points over $[t, t + 1]$ and roughly 8 basis points or more over $[t, t + 10]$ in the cleaner Pattern-firm design.
- **Dynamics.** The price effect accumulates over one to two weeks, peaks around 10 basis points, and then reverses over later weeks. The reversal supports temporary attention-driven price pressure rather than permanent information spillovers.
- **Trading mechanism.** Retail order imbalance rises after the cue. The time path of retail buying closely matches the time path of returns: buying builds up with returns and selling coincides with reversal.
- **Memory-theory tests.** The effect is stronger when memory theory predicts a stronger association: greater attention during encoding, more similarity between encoded experiences, less interference from other announcements, and more surprising cues.
- **Cross-sectional evidence.** Large cueing firms and small cued firms drive much of the effect. This matches the attention mechanism: large announcements are more visible, and small stocks are easier for retail order flow to move.
- **Trading-strategy result.** A long-short strategy based on cues generates significant alpha in 2015–2019, and the alpha is driven by the long leg, consistent with buying pressure rather than risk exposure.
- **Economic takeaway.** Memory moves markets because investors' recall processes aggregate into trades. Context-based memory associations can become a market-level force that creates temporary return predictability.

2 Data and measurement

2.1 Sample construction

- The main sample spans 2005–2019, the period in which the paper has data on Pattern firms.
- The unit is a firm-day for which characteristic-adjusted returns can be calculated over $[t, t + 1]$ and $[t, t + 10]$.
- Days are trading days. Returns are characteristic-adjusted returns rather than raw returns.
- Characteristic-adjusted returns are formed by matching each stock to one of 125 portfolios sorted by size, book-to-market, and momentum, then subtracting the value-weighted return of the matched portfolio.
- Market capitalization three trading days before the return date is used as the portfolio weight.

Detailed interpretation. The return measure is designed to remove standard size, value, and momentum effects. This is important because the cue variable is rare and the effect size is measured in basis points.

2.2 Pattern firms and calendar rotations

- Pattern firms are firms that repeatedly announce earnings according to a rigid calendar rule, such as the first Thursday of a fiscal-quarter month.
- Calendar rotations change the weekday on which a month starts, shifting rigid announcement dates mechanically from one calendar date to another.
- These rotations create earnings-date overlaps that are plausibly unrelated to firms' fundamentals.
- The paper uses those mechanically shifted overlaps to identify which firms were experienced in the same context and therefore encoded as associated in memory.

Detailed interpretation. Pattern firms and calendar rotations solve the key identification problem: memory associations need to be observed without using discretionary announcement timing that may itself reveal fundamentals.

2.3 Memory cue variable

For firm i on day t , define

$$Cue_{i,t} = 1$$

if at least one firm that announced earnings on the same day as firm i in the previous fiscal quarter announces earnings on day t .

- The cueing firm is the firm announcing earnings on day t .
- The cued firm is the associated firm that does not announce earnings on day t .
- The baseline effect focuses on cues from large cueing firms because large firms' announcements are more visible.
- On firm-days with multiple cues, the paper also measures the number of cues and the surprise of the cueing firms.

Detailed interpretation. The cue variable captures a pure recall event: attention to the cueing firm may retrieve the cued firm from memory even though the cued firm has no own announcement.

2.4 Retail order imbalance

The paper uses TAQ data to infer retail trading and constructs retail order imbalance:

$$RetailOI_{i,t} = \frac{RetailBuyVol_{i,t} - RetailSellVol_{i,t}}{RetailBuyVol_{i,t} + RetailSellVol_{i,t}}.$$

- The baseline retail classification uses the Barber et al. (2024) BHJOS algorithm.
- The paper also uses the Boehmer et al. (2021) BJZZ algorithm.
- BJZZ retail order imbalance is computed using all lots and, separately, only odd-lot trades.
- Odd lots are more likely to be retail trades, so the odd-lot version is a useful robustness check.

Detailed interpretation. Retail order imbalance provides the mechanism test. If cues create attention and retail investors are net buyers of attention-grabbing stocks, then cues should generate positive retail order imbalance.

2.5 Controls and exclusions

- The baseline excludes firm pairs in the same TNIC-2 industry.
- It excludes vertically related firms using the VTNIC database.
- It excludes firms with overlapping analyst coverage.
- Regressions include own earnings-announcement-window dummies, own 8-K filings, Capital IQ Key Developments, RavenPack positive and negative news indicators, and news-volume changes.
- Firm-by-year-quarter fixed effects absorb persistent firm-level fundamentals within a fiscal quarter.
- Standard errors are clustered by firm and trading day.

Detailed interpretation. These restrictions and controls are designed to ensure that the cue is not merely an information spillover from a fundamentally related firm.

3 Models and empirical specifications

3.1 Baseline return regression

$$Return_i[t, t+h] = \alpha + \beta Cue_{i,t} + Controls_{i,t} + FE_{i,yq} + \varepsilon_{i,t}.$$

Detailed interpretation. The coefficient β is the abnormal return of the cued firm after a memory cue. The main windows are $[t, t+1]$ and $[t, t+10]$. A positive coefficient means that the cue produces price pressure in the associated firm's stock.

3.2 Event-study dynamics

$$Return_i[t, t+k] = \alpha_k + \beta_k Cue_{i,t} + Controls_{i,t} + FE_{i,yq} + \varepsilon_{i,t}, \quad k = 1, \dots, 50.$$

Detailed interpretation. The sequence β_k traces whether the effect builds, persists, or reverses. The paper's return dynamics are central because a reversal is hard to reconcile with permanent fundamental information and much easier to reconcile with temporary attention-induced buying pressure.

3.3 Retail order imbalance regression

$$RetailOI_i[t, t+h] = \alpha + \beta Cue_{i,t} + Controls_{i,t} + FE_{i,yq} + \varepsilon_{i,t}.$$

Detailed interpretation. Positive coefficients in the retail OI regressions indicate net retail buying after memory cues. This links the cognitive mechanism to actual trading behavior.

3.4 Cross-sectional memory-strength tests

The paper estimates variants of

$$Return_i[t, t+10] = \alpha + \beta_1 CueStrong_{i,t} + \beta_2 CueOther_{i,t} + Controls + FE + \varepsilon_{i,t}.$$

Detailed interpretation. These specifications ask whether the effect is stronger when memory theory predicts stronger recall: more attention during encoding, more similarity of the encoded experience, less interference, and more salient cueing events.

3.5 Trading strategy regressions

$$R_t^{strategy} = \alpha + b_M MKT_t + b_S SMB_t + b_H HML_t + b_{Mom} MOM_t + b_{Rev} STRev_t + \varepsilon_t.$$

Detailed interpretation. The trading strategy is not presented as a practical arbitrage after transaction costs. It is a calendar-time asset-pricing robustness check that asks whether cue-based return predictability remains after standard risk-factor adjustment.

4 Identification and empirical design

4.1 Ideal experiment

The ideal experiment would randomly assign firms to be associated in investors' memory and later randomly cue one member of the pair. The paper cannot literally manipulate investor memory, so it builds a quasi-experiment from earnings-announcement calendar mechanics.

4.2 Why calendar rotations help

- Pattern-firm schedules are rigid, such as first Thursday or fifth Thursday rules.
- Calendar rotations shift the dates on which those rules fall.
- Two firms can overlap in one quarter due to the calendar but not overlap in the next quarter.
- The overlap defines a memory encoding event, and the later announcement defines a cue.

Detailed interpretation. The identifying variation comes from the calendar, not from discretionary managerial timing. This is the reason the paper focuses on Pattern firms.

4.3 Main threat: fundamental relatedness

The key concern is that firms that announce together may be fundamentally related. The paper addresses this by excluding:

- same TNIC-2 industry pairs;
- vertically related pairs;
- pairs with overlapping analyst coverage.

Detailed interpretation. These exclusions push the design toward contextual memory rather than economic similarity. The paper is not simply showing that related firms move together after news.

4.4 News controls and own events

The paper controls for a wide set of own-firm events around the cue date:

- own earnings announcements over $[t - 10, t + 10]$;
- own 8-K filings;
- own Capital IQ Key Developments;
- positive and negative RavenPack news;
- log changes in news volume.

Detailed interpretation. These controls make it less likely that the cue variable captures ordinary firm-specific news or concurrent attention shocks unrelated to memory.

4.5 Mechanism validation

The identification argument is strengthened by several mechanism tests:

- retail order imbalance rises after cues;
- the return effect builds and then reverses;
- large cueing firms and small cued firms drive the effect;
- stronger encoding attention increases the effect;
- similarity of experiences modulates the effect;
- interference weakens the effect;
- anticipation and surprise tests are inconsistent with a simple information-spillover story.

Detailed interpretation. The paper’s credibility comes from the full pattern. It is not just one positive return coefficient; many comparative statics line up with associative memory theory.

5 Section-by-section study guide

- **Introduction.** Introduces internally generated attention and distinguishes memory from external attention sources.
- **Empirical strategy.** Defines Pattern firms, calendar rotations, memory encoding, and cues.
- **Data.** Describes characteristic-adjusted returns, retail order imbalance, and summary statistics.
- **Main return results.** Shows cue-induced abnormal returns in the clean Pattern-firm sample and explains why the all-EA sample is contaminated by fundamentals.
- **Return dynamics.** Shows buildup and reversal of the effect.
- **Retail trading.** Establishes that retail order imbalance responds to cues and mirrors return dynamics.
- **Psychological mechanism.** Tests attention at encoding, similarity, interference, alternative cues, anticipation, and surprise.
- **Trading strategy.** Converts cue predictability into a calendar-time long-short portfolio test.

6 Tables and figures

6.1 Figure 1: Calendar rotations - an example + Table 1: Summary statistics

Read:

- Figure 1 shows how two firms can announce earnings on the same date in one quarter but on different dates in the next quarter because the month begins on a different weekday.
- This is the paper’s visual explanation of how calendar rotations create plausibly exogenous memory encoding events.
- Table 1 shows that cue events are rare: $Cue = 1$ on 0.64% of firm-days.
- There are 105,550 cue firm-days, and the average number of cues on a cue day is 1.893.
- The return sample contains 16,502,158 firm-days, and retail order imbalance samples are smaller because TAQ-based retail measures begin later.
- Retail order imbalance is slightly negative on average, so the mechanism tests are not mechanically driven by a general retail-buying bias.

Detailed interpretation. Figure 1 clarifies why the empirical design is not simply exploiting firms that habitually announce together. The same rigid rule generates overlap in one quarter and non-overlap in another quarter because the calendar changes. Table 1 then shows the scale of the resulting design. The cue is rare enough that it is unlikely to reflect a broad market-wide event, but common enough to support high-powered panel regressions.

Economic message. The empirical design turns a small, mechanically generated attention event into a measurable market-level treatment.

June, July, August 2013						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
30	1	2	3	4	5	6
Fiscal quarter end (June 30th)	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31	1	2	3
			A and B announce			
4	5	6	7	8	9	10

June, July, August 2014						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
29	30	1	2	3	4	5
Fiscal quarter end (June 30th)	7	8	9	10	11	12
6						
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
				B announces		
3	4	5	6	7	8	9
			A announces			

Figure 1
Calendar rotations: An example.

(a) Figure 1: Calendar rotations - an example

Table 1 Summary statistics							
	Mean	p25	p50	p75	Std. Dev.	Min	Max
Return in % over $[t, t+1]$	-0.0110	-1.6360	-0.0844	1.4695	3.6471	-12.1098	15.7098
Return in % over $[t, t+10]$	-0.1121	-4.1007	-0.2875	3.3323	6.4877	-26.1222	31.6051
BIROS retail OI in % over $[t, t+1]$	-4.4383	-19.8704	-2.4291	11.8036	30.9262	-100.0000	84.8624
BIROS retail OI in % over $[t, t+10]$	-3.9705	-11.8761	-2.7535	6.9529	14.5156	-47.8004	35.3404
BZZ retail OI in % over $[t, t+1]$	-2.6942	-18.4402	-1.9388	13.1443	26.4661	-71.7819	65.7579
BZZ retail OI in % over $[t, t+10]$	-2.5887	-10.1575	-1.8349	5.4210	12.8316	-38.4445	29.1627
BZZ retail OI in % (odd lots) over $[t, t+1]$	-0.3178	-15.8640	0.2656	15.5155	27.2288	-73.4037	70.9338
BZZ retail OI in % (odd lots) over $[t, t+10]$	-0.1007	-4.2258	0.2125	6.2355	13.6862	-37.4915	34.7560
Cue (dummy)	0.0064	0.0000	0.0000	0.0000	0.0797	0.0000	1.0000
Number of cues	1.8932	1	0.0000	0.0000	1.7852	0	13
Surprise of cue (EW)	0.0005	0.0000	0.0003	0.0000	0.0014	-0.0040	0.0069
Surprise of cue (VW)	0.0005	0.0000	0.0002	0.0010	0.0014	-0.0042	0.0068

This table contains summary statistics of the sample used in the empirical analysis. The sample covers years 2005–2019. Return over the time window $[t, t+1]$ is the raw return of a stock over the time window $[t, t+1]$ minus the value-weighted return of a portfolio of stocks matched on size, book-to-market, and momentum. Retail order imbalance (OI) is calculated daily from Trade and Quote (TAQ) data using either the Barber et al. (2006) (BIROS) or the Boehmer et al. (2011) (BZZ) algorithm. Retail order imbalance over the time window $[t, t+1]$ is the average order imbalance over that time window. Data on retail order imbalance are available for years 2007–2019, and for years 2013–2019 when constructed using only odd lot trades (trades of fewer than 100 shares). Cue is a dummy variable equal to one if at least one firm that announced earnings on the same day as firm i in the previous fiscal quarter announced earnings on day t . Number of cues is the number of issuing firms announcing on firm-days with Cue equal to one. Surprise is the earnings surprise of the issuing firm(s) announcing on firm-days with Cue equal to one. On firm-days with multiple cues, I calculate both the equally-weighted (EW) and value-weighted (VW) earnings surprise.

(b) Table 1: Summary statistics

6.2 Table 2: Return results

Read:

- Panel A is the clean Pattern-firm design. It estimates returns for cued firms when associations are created by exogenous calendar rotations.
- The no-control specification shows a roughly 6.2 bps abnormal return over $[t, t + 1]$.
- With controls and firm-year-quarter fixed effects, the effect remains positive in the short window and is larger over $[t, t + 10]$.
- Panel B uses all earnings announcements from 1995–2020. The effect appears in simple regressions but fades once rich controls and firm-year-quarter fixed effects are added.
- The contrast between Panels A and B is central: the clean Pattern-firm design survives controls, while the broader all-EA design is contaminated by fundamentals and strategic announcement timing.

Detailed interpretation. Table 2 is the paper’s core return table. Its most important lesson is not just that the cue coefficient is positive, but that identification quality changes the result. The Pattern-firm sample isolates memory cues through calendar rotations. The all-EA sample contains endogenous earnings-announcement timing and fundamental information spillovers, so controls absorb the apparent effect. This is exactly why the paper is built around Pattern firms.

Economic message. Memory-induced attention creates temporary positive abnormal returns, but clean identification is essential for separating memory from information spillovers.

Table 2
Return results
A. Main sample

Dependent variable:	Return $[t, t+1]$ in %						Return $[t, t+10]$ in %					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Cue $[t]$	0.062*** (0.016)	0.063*** (0.016)	0.062*** (0.016)	0.043*** (0.017)	0.038*** (0.016)	0.018 (0.040)	0.171*** (0.040)	0.180*** (0.040)	0.178*** (0.040)	0.146*** (0.034)	0.084*** (0.034)	0.046 (0.024)
Ow EA $[t-10]$	0.009 (0.010)	0.010 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)	0.006 (0.010)	0.014 (0.024)	0.014 (0.024)	0.014 (0.024)	0.014 (0.022)	0.014 (0.022)	0.002 (0.021)
Ow EA $[t-9]$	0.014 (0.011)	0.013 (0.011)	0.010 (0.011)	0.012 (0.011)	0.012 (0.011)	0.008 (0.011)	0.012 (0.024)	0.012 (0.024)	0.012 (0.024)	0.009 (0.022)	0.012 (0.022)	0.007*** (0.021)
Ow EA $[t-8]$	0.016 (0.010)	0.016 (0.010)	0.016 (0.010)	0.015 (0.010)	0.015 (0.010)	0.014 (0.010)	0.012 (0.024)	0.012 (0.024)	0.012 (0.024)	0.009 (0.022)	0.012 (0.022)	0.007*** (0.021)
Ow EA $[t-7]$	0.021*** (0.010)	0.021*** (0.010)	0.021*** (0.010)	0.019*** (0.010)	0.022*** (0.010)	0.022*** (0.010)	0.013 (0.024)	0.013 (0.024)	0.013 (0.024)	0.012 (0.022)	0.012 (0.022)	0.011*** (0.021)
Ow EA $[t-6]$	0.012 (0.010)	0.012 (0.010)	0.014 (0.010)	0.011 (0.010)	0.011 (0.010)	0.011 (0.010)	0.041*** (0.024)	0.041*** (0.024)	0.041*** (0.024)	0.039*** (0.022)	0.041*** (0.022)	0.038*** (0.021)
Ow EA $[t-5]$	0.023*** (0.011)	0.022*** (0.011)	0.022*** (0.011)	0.021*** (0.011)	0.021*** (0.011)	0.015 (0.011)	0.054*** (0.025)	0.054*** (0.025)	0.052*** (0.025)	0.042*** (0.024)	0.042*** (0.024)	0.039*** (0.023)
Ow EA $[t-4]$	0.037*** (0.011)	0.037*** (0.011)	0.037*** (0.011)	0.034*** (0.011)	0.034*** (0.011)	0.027*** (0.011)	0.084*** (0.025)	0.084*** (0.025)	0.082*** (0.025)	0.069*** (0.024)	0.069*** (0.024)	0.057*** (0.023)
Ow EA $[t-3]$	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.011 (0.010)	0.011 (0.010)	0.007 (0.010)	0.079*** (0.025)	0.079*** (0.025)	0.079*** (0.025)	0.063*** (0.024)	0.063*** (0.024)	0.056*** (0.023)
Ow EA $[t-2]$	0.011 (0.012)	0.011 (0.012)	0.011 (0.012)	0.011 (0.012)	0.011 (0.012)	0.007 (0.012)	0.079*** (0.025)	0.079*** (0.025)	0.079*** (0.025)	0.063*** (0.024)	0.063*** (0.024)	0.056*** (0.023)
Ow EA $[t-1]$	-0.159*** (0.015)	-0.163*** (0.015)	-0.163*** (0.015)	-0.164*** (0.015)	-0.164*** (0.015)	-0.164*** (0.015)	-0.164*** (0.029)	-0.164*** (0.029)	-0.164*** (0.029)	-0.164*** (0.028)	-0.164*** (0.028)	-0.164*** (0.028)
Ow EA $[t]$	-0.048*** (0.019)	-0.124*** (0.020)	-0.124*** (0.020)	-0.124*** (0.020)	-0.124*** (0.020)	-0.124*** (0.020)	-0.124*** (0.020)	-0.124*** (0.020)	-0.124*** (0.020)	-0.124*** (0.020)	-0.124*** (0.020)	-0.124*** (0.020)
Ow EA $[t+1]$	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)	0.017 (0.017)
Ow EA $[t+2]$	0.148*** (0.012)	0.141*** (0.012)	0.141*** (0.012)	0.141*** (0.012)	0.141*** (0.012)	0.141*** (0.012)	0.141*** (0.012)	0.141*** (0.012)	0.141*** (0.012)	0.141*** (0.012)	0.141*** (0.012)	0.141*** (0.012)
Ow EA $[t+3]$	0.073*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)

(a) Table 2, Panel A: Pattern-firm return results, part 1

Table 2
Continued
A. Main sample

Dependent variable:	Return $[t, t+1]$ in %						Return $[t, t+10]$ in %					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Ow EA $[t+4]$	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)	0.045*** (0.011)
Ow EA $[t+5]$	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)
Ow EA $[t+6]$	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)	0.041*** (0.010)
Ow EA $[t+7]$	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)	0.039*** (0.010)
Ow EA $[t+8]$	0.010 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)	0.009 (0.010)
Ow EA $[t+9]$	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)	-0.011 (0.010)
Ow EA $[t+10]$	-0.008 (0.010)	-0.013 (0.010)	-0.013 (0.010)	-0.013 (0.010)	-0.013 (0.010)	-0.013 (0.010)	-0.013 (0.010)	-0.013 (0.010)	-0.013 (0.010)	-0.013 (0.010)	-0.013 (0.010)	-0.013 (0.010)
Ow S-K $[t]$	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)
Ow key development $[t]$	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)	0.007 (0.007)
Positive news coverage $[t]$	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)
Negative news coverage $[t]$	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)	0.009 (0.011)
log $t + 1$ increase in news volume	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)	0.127*** (0.007)
log $t + 1$ decrease in news volume	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)	0.115*** (0.006)
Day FE	no	no	no	no	no	no	yes	yes	yes	yes	yes	yes
Firm x Year-quarter FE	no	no	no	no	no	no	yes	yes	yes	yes	yes	yes
Observations	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158
R-squared	.000	.000	.001	.001	.001	.001	.001	.001	.001	.001	.001	.001

(b) Table 2, Panel A: controls and longer-window results

Table 2
Continued
B. Using all EAs

Dependent variable:	Return $[t, t+1]$ in %											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Cue $[t]$	0.050*** (0.006)	0.028*** (0.006)	0.022*** (0.006)	0.016*** (0.006)	0.007 (0.005)	-0.002 (0.005)	0.089*** (0.018)	0.059*** (0.017)	0.060*** (0.017)	0.055*** (0.016)	-0.018 (0.015)	-0.011 (0.014)
Own EA $[t-10]$	0.021*** (0.008)	0.021*** (0.008)	0.023*** (0.008)	0.019*** (0.008)	0.018*** (0.008)	0.018*** (0.008)	0.099*** (0.020)	0.095*** (0.020)	0.149*** (0.019)	0.064*** (0.018)	0.188*** (0.017)	0.188*** (0.017)
Own EA $[t-9]$	0.012 (0.009)	0.015* (0.009)	0.015* (0.009)	0.009 (0.008)	0.010 (0.008)	0.010 (0.008)	0.094*** (0.021)	0.094*** (0.021)	0.154*** (0.020)	0.111*** (0.019)	0.111*** (0.018)	0.111*** (0.018)
Own EA $[t-8]$	0.018** (0.008)	0.018** (0.008)	0.022*** (0.008)	0.015* (0.008)	0.016** (0.008)	0.016** (0.008)	0.094*** (0.021)	0.093*** (0.021)	0.152*** (0.020)	0.103*** (0.019)	0.103*** (0.019)	0.103*** (0.019)
Own EA $[t-7]$	0.024*** (0.008)	0.024*** (0.008)	0.027*** (0.008)	0.021*** (0.008)	0.022*** (0.008)	0.022*** (0.008)	0.083*** (0.021)	0.083*** (0.021)	0.143*** (0.020)	0.099*** (0.019)	0.099*** (0.019)	0.099*** (0.019)
Own EA $[t-6]$	0.007 (0.009)	0.007 (0.009)	0.009 (0.009)	0.005 (0.008)	0.005 (0.008)	0.005 (0.008)	0.078*** (0.021)	0.078*** (0.021)	0.132*** (0.020)	0.049** (0.019)	0.049** (0.019)	0.049** (0.019)
Own EA $[t-5]$	0.006 (0.009)	0.005 (0.009)	0.007 (0.009)	0.004 (0.008)	0.003 (0.008)	0.003 (0.008)	0.081*** (0.021)	0.081*** (0.021)	0.135*** (0.020)	0.099*** (0.019)	0.099*** (0.019)	0.099*** (0.019)
Own EA $[t-4]$	0.001 (0.009)	-0.000 (0.009)	0.006 (0.008)	-0.002 (0.008)	0.001 (0.008)	0.001 (0.008)	0.077*** (0.021)	0.076*** (0.021)	0.137*** (0.020)	0.080*** (0.019)	0.080*** (0.019)	0.080*** (0.019)
Own EA $[t-3]$	-0.024*** (0.009)	-0.023*** (0.009)	-0.017** (0.009)	-0.026*** (0.009)	-0.022*** (0.009)	-0.022*** (0.009)	0.083*** (0.021)	0.083*** (0.021)	0.105*** (0.020)	0.099*** (0.019)	0.099*** (0.019)	0.099*** (0.019)
Own EA $[t-2]$	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.010 (0.010)	0.009 (0.009)	0.009 (0.009)	0.021 (0.022)	0.021 (0.022)	0.020 (0.021)	0.020 (0.021)	0.021 (0.021)	0.021 (0.021)
Own EA $[t-1]$	-0.162*** (0.012)	-0.162*** (0.012)	-0.158*** (0.012)	-0.163*** (0.011)	-0.163*** (0.011)	-0.163*** (0.011)	-0.124*** (0.024)	-0.124*** (0.024)	-0.166*** (0.023)	-0.146*** (0.023)	-0.139*** (0.024)	-0.139*** (0.024)
Own EA $[t]$	-0.002 (0.012)	-0.048*** (0.012)	-0.047*** (0.012)	-0.052*** (0.012)	-0.054*** (0.012)	-0.054*** (0.012)	-0.066*** (0.028)	-0.066*** (0.028)	-0.117*** (0.029)	-0.110*** (0.028)	-0.110*** (0.027)	-0.110*** (0.027)
Own EA $[t+1]$	0.237*** (0.013)	0.206*** (0.013)	0.212*** (0.013)	0.204*** (0.013)	0.205*** (0.013)	0.205*** (0.013)	0.082*** (0.027)	0.082*** (0.027)	0.079*** (0.025)	0.051 (0.025)	0.043 (0.026)	0.043 (0.026)
Own EA $[t+2]$	0.238*** (0.010)	0.235*** (0.010)	0.242*** (0.011)	0.231*** (0.010)	0.236*** (0.010)	0.236*** (0.010)	0.160*** (0.027)	0.159*** (0.027)	0.189*** (0.026)	0.131*** (0.025)	0.144*** (0.026)	0.144*** (0.026)
Own EA $[t+3]$	0.144*** (0.009)	0.142*** (0.009)	0.144*** (0.009)	0.139*** (0.009)	0.135*** (0.009)	0.135*** (0.009)	0.203*** (0.027)	0.200*** (0.027)	0.226*** (0.026)	0.172*** (0.024)	0.172*** (0.024)	0.172*** (0.024)
Own EA $[t+4]$	0.093*** (0.009)	0.090*** (0.009)	0.091*** (0.009)	0.088*** (0.009)	0.075*** (0.009)	0.075*** (0.009)	0.223*** (0.027)	0.220*** (0.027)	0.239*** (0.026)	0.194*** (0.025)	0.194*** (0.025)	0.194*** (0.025)

(Continued)

Table 2
Continued
B. Using all EAs

Dependent variable:	Return $[t, t+1]$ in %											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Own EA $[t+5]$	0.062*** (0.009)	0.057*** (0.009)	0.057*** (0.009)	0.049*** (0.009)	0.055*** (0.009)	0.043*** (0.009)	0.244*** (0.027)	0.237*** (0.027)	0.246*** (0.028)	0.214*** (0.025)	0.200*** (0.025)	0.200*** (0.025)
Own EA $[t+6]$	0.054*** (0.009)	0.051*** (0.009)	0.049*** (0.009)	0.048*** (0.008)	0.041*** (0.008)	0.041*** (0.008)	0.261*** (0.027)	0.258*** (0.027)	0.260*** (0.028)	0.238*** (0.024)	0.209*** (0.025)	0.209*** (0.025)
Own EA $[t+7]$	0.039*** (0.009)	0.039*** (0.009)	0.039*** (0.009)	0.035*** (0.008)	0.039*** (0.008)	0.039*** (0.008)	0.275*** (0.027)	0.273*** (0.027)	0.266*** (0.028)	0.240*** (0.024)	0.211*** (0.025)	0.211*** (0.025)
Own EA $[t+8]$	0.029*** (0.009)	0.027*** (0.009)	0.028*** (0.009)	0.024*** (0.008)	0.013 (0.008)	0.013 (0.008)	0.305*** (0.027)	0.304*** (0.027)	0.287*** (0.028)	0.272*** (0.024)	0.233*** (0.025)	0.233*** (0.025)
Own EA $[t+9]$	0.016* (0.008)	0.014* (0.008)	-0.000 (0.008)	0.010 (0.008)	-0.008 (0.008)	-0.008 (0.008)	0.360*** (0.027)	0.358*** (0.027)	0.324*** (0.028)	0.324*** (0.024)	0.277*** (0.025)	0.277*** (0.025)
Own EA $[t+10]$	0.016* (0.009)	0.012 (0.009)	-0.001 (0.009)	0.007 (0.008)	-0.009 (0.008)	-0.009 (0.008)	0.478*** (0.025)	0.473*** (0.025)	0.454*** (0.026)	0.440*** (0.024)	0.382*** (0.025)	0.382*** (0.025)
Own 8-K $[t]$	-0.074*** (0.007)	-0.074*** (0.007)	-0.067*** (0.007)	-0.066*** (0.007)	-0.134*** (0.016)	-0.134*** (0.015)	-0.075*** (0.021)	-0.075*** (0.021)	-0.075*** (0.021)	-0.075*** (0.019)	-0.075*** (0.019)	-0.075*** (0.019)
Own key development $[t]$	0.007 (0.005)	0.007 (0.005)	0.007 (0.005)	0.007 (0.004)	0.007 (0.004)	0.007 (0.004)	0.121*** (0.025)	0.122*** (0.025)	0.122*** (0.024)	0.125*** (0.024)	0.109*** (0.020)	0.109*** (0.020)
Positive news coverage $[t]$	0.11*** (0.009)	0.11*** (0.009)	0.11*** (0.009)	0.107*** (0.008)	0.107*** (0.008)	0.107*** (0.008)	-0.119*** (0.011)	-0.119*** (0.011)	-0.109*** (0.010)	-0.109*** (0.010)	-0.109*** (0.011)	-0.109*** (0.011)
Negative news coverage $[t]$	-0.119*** (0.011)	-0.119*** (0.011)	-0.119*** (0.011)	-0.107*** (0.010)	-0.107*** (0.010)	-0.107*** (0.010)	0.123*** (0.008)	0.123*** (0.008)	0.123*** (0.007)	0.123*** (0.007)	0.123*** (0.007)	0.123*** (0.007)
log(1 + increase in news volume)	0.137*** (0.006)	0.137*** (0.006)	0.137*** (0.006)	0.134*** (0.006)	0.134*** (0.006)	0.134*** (0.006)	0.219*** (0.013)	0.219*** (0.013)	0.219*** (0.013)	0.219*** (0.010)	0.219*** (0.010)	0.219*** (0.010)
log(1 + decrease in news volume)	0.137*** (0.006)	0.137*** (0.006)	0.137*** (0.006)	0.134*** (0.006)	0.134*** (0.006)	0.134*** (0.006)	0.219*** (0.013)	0.219*** (0.013)	0.219*** (0.013)	0.219*** (0.010)	0.219*** (0.010)	0.219*** (0.010)
Day FE	no	no	no	no	yes	yes	no	no	no	yes	yes	yes
Firm x Year-quarter FE	no	no	no	no	yes	yes	no	no	no	yes	yes	yes
Observations	32,631,572	32,631,572	32,631,572	32,631,572	32,631,572	32,631,572	32,631,572	32,631,572	32,631,572	32,631,572	32,631,572	32,631,572
R-squared	.000	.000	.001	.005	.026	.029	.000	.000	.000	.005	.151	.154

This table shows the return effect of a memory cue. In column 1, the return over $[t, t+1]$ is regressed on the cue dummy without any further controls. Column 2 adds 21 dummy variables that capture a potential own earnings announcement in the window $[t-10, t+10]$. Column 3 adds dummy variables that capture the timing of an own 8-K, an own Key Development, and positive/negative news coverage. This column also controls for the logarithm of 1 plus the absolute increase or decrease in the volume of news coverage. Columns 4 and 5 add day fixed effects and firm x year-quarter fixed effects, respectively. Column 6 adds both day and firm x year-quarter fixed effects simultaneously. Columns 7-12 mirror columns 1-6, except that the dependent variable is the return over $[t, t+10]$. Panel A shows results for the main sample, in which memory associations are estimated using only earnings announcements that are exogenously dated by calendar releases. Panel B shows results for a broader sample, in which memory associations are estimated using all earnings announcements of all firms. The sample in panel A covers years 2005-2019 and the sample in panel B covers years 1995-2020. Standard errors are clustered by firm and trading day and are displayed in parentheses below the coefficients. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 2, Panel B: all-earnings-announcement sample, part 1

(b) Table 2, Panel B: all-earnings-announcement sample, part 2

6.3 Figure 2: Return dynamics and long-run reversal

Read:

- Figure 2 estimates 50 expanding-window return regressions, from $[t, t+1]$ to $[t, t+50]$.
- The cue effect accumulates over the first one to two weeks.
- The cumulative return peaks around 10 basis points.
- After the peak, the cumulative return reverses and approaches zero by roughly $t+25$ to $t+30$.
- The right panel plots residuals after subtracting the Table 2 short-window and long-window effects, making the reversal easier to see.

Detailed interpretation. The reversal is the crucial difference between memory-induced price pressure and fundamental information. If the cue revealed permanent information about the cued firm, the effect should persist. Instead, the stock temporarily appreciates as attention-driven buying accumulates and then reverses when the buying pressure fades or is arbitrated away.

Economic message. The figure shows a transitory mispricing pattern: memory cues move prices, but the market later corrects the pressure.

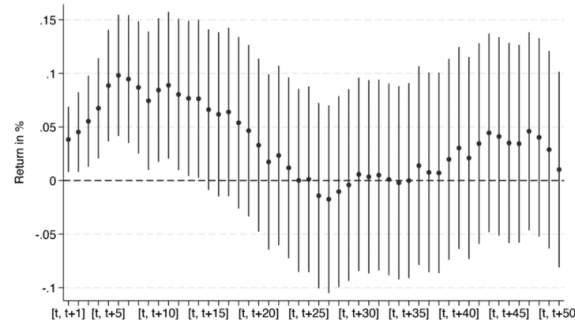


Figure 2
Return dynamics and long-run reversal
This figure traces out the return dynamics in response to a memory cue by showing results from 50 separate regressions. The dependent variable in each regression is the cumulative return over a different return window, ranging from $[t, t+1]$ to $[t, t+50]$. The dots in the figure represent the coefficients on the cue dummy, along with 95% confidence intervals, from each regression. All regressions include the full set of controls from Table 2 and firm-year-quarter fixed effects. Standard errors are clustered by firm and trading day.

Figure 2: Return dynamics and long-run reversal

6.4 Table 3: Retail order imbalance results + Figure 3: Retail order imbalance dynamics

Read:

- Table 3 uses three retail-trade measures: BHJOS, BJZZ all lots, and BJZZ odd lots.
- For BHJOS, the $[t, t + 10]$ cue coefficient is 0.220 and significant, while the $[t, t + 1]$ coefficient is positive but not significant.
- For BJZZ all lots, the $[t, t + 10]$ coefficient is 0.118 and significant at the 10% level.
- For BJZZ odd lots, the $[t, t + 10]$ coefficient is 0.329 and significant, stronger than the all-lots BJZZ estimate.
- Figure 3 shows that retail OI rises after a cue, peaks around the same period as returns, and falls when returns reverse.

Detailed interpretation. Table 3 is the mechanism table: it translates price pressure into actual investor behavior. The stronger odd-lot result is important because odd lots are more likely to be retail trades. Figure 3 then shows timing: the trading pressure does not appear at a random horizon but accumulates exactly when returns build and fades when returns reverse. This strengthens the interpretation that retail attention and buying pressure cause the return pattern.

Economic message. Memory cues do not merely correlate with returns; they generate retail buying pressure with matching dynamics.

Table 3
Retail order imbalance results

Dependent variable:	BHJOS retail OI in %		BJZZ retail OI in % (all lots)		BJZZ retail OI in % (odd lots)	
Window:	$[t, t+1]$ (1)	$[t, t+10]$ (2)	$[t, t+1]$ (3)	$[t, t+10]$ (4)	$[t, t+1]$ (5)	$[t, t+10]$ (6)
Cue $[r]$	0.179 (0.140)	0.220*** (0.070)	0.031 (0.116)	0.118* (0.064)	0.280 (0.185)	0.329*** (0.096)
Own EA $[r - 10]$	0.173 (0.110)	0.096* (0.057)	0.091 (0.099)	-0.039 (0.051)	-0.219 (0.141)	-0.175** (0.071)
Own EA $[r - 9]$	0.274** (0.112)	0.094 (0.058)	0.123 (0.097)	-0.036 (0.052)	0.013 (0.144)	-0.105 (0.073)
Own EA $[r - 8]$	0.428*** (0.111)	0.110* (0.059)	0.143 (0.097)	-0.036 (0.053)	0.018 (0.143)	-0.114 (0.074)
Own EA $[r - 7]$	0.330*** (0.111)	0.134** (0.060)	0.116 (0.096)	-0.018 (0.053)	-0.002 (0.135)	-0.115 (0.077)
Own EA $[r - 6]$	0.353*** (0.109)	0.136** (0.060)	0.129 (0.095)	-0.017 (0.054)	-0.100 (0.141)	-0.112 (0.079)
Own EA $[r - 5]$	0.458*** (0.113)	0.156*** (0.060)	0.127 (0.100)	-0.031 (0.054)	-0.006 (0.144)	-0.121 (0.079)
Own EA $[r - 4]$	0.593*** (0.112)	0.200*** (0.061)	0.194** (0.097)	-0.014 (0.054)	0.193 (0.136)	-0.070 (0.080)
Own EA $[r - 3]$	0.671*** (0.111)	0.223*** (0.061)	0.193** (0.097)	-0.022 (0.055)	-0.045 (0.138)	-0.081 (0.081)
Own EA $[r - 2]$	0.474*** (0.110)	0.228*** (0.062)	-0.021 (0.097)	-0.034 (0.056)	-0.277* (0.149)	-0.077 (0.082)
Own EA $[r - 1]$	0.491*** (0.111)	0.282*** (0.063)	-0.231** (0.100)	-0.041 (0.056)	-0.752*** (0.151)	-0.161* (0.083)
Own EA $[r]$	0.312*** (0.116)	0.250*** (0.064)	-0.487*** (0.100)	-0.065 (0.057)	-1.202*** (0.171)	-0.279*** (0.084)
Own EA $[r + 1]$	3.665*** (0.114)	0.894*** (0.061)	2.677*** (0.102)	0.488*** (0.054)	2.661*** (0.166)	0.334*** (0.083)
Own EA $[r + 2]$	5.024*** (0.128)	1.199*** (0.060)	4.352*** (0.117)	0.741*** (0.053)	4.290*** (0.176)	0.582*** (0.081)
Own EA $[r + 3]$	2.352*** (0.119)	1.325*** (0.059)	2.203*** (0.106)	0.894*** (0.053)	1.744*** (0.162)	0.718*** (0.081)
Own EA $[r + 4]$	1.575*** (0.114)	1.421*** (0.059)	1.435*** (0.102)	0.991*** (0.053)	1.257*** (0.163)	0.825*** (0.083)
Own EA $[r + 5]$	1.074*** (0.112)	1.460*** (0.059)	1.045*** (0.102)	1.050*** (0.053)	0.899*** (0.151)	0.875*** (0.083)
Own EA $[r + 6]$	0.753*** (0.114)	1.491*** (0.058)	0.828*** (0.098)	1.122*** (0.053)	0.608*** (0.152)	0.888*** (0.083)
Own EA $[r + 7]$	0.685*** (0.111)	1.501*** (0.059)	0.814*** (0.101)	1.179*** (0.053)	0.413*** (0.156)	0.901*** (0.083)

(Continued)

Table 3
Continued

Dependent variable:	BHJOS retail OI in %		BJZZ retail OI in % (all lots)		BJZZ retail OI in % (odd lots)	
Window:	$[t, t+1]$ (1)	$[t, t+10]$ (2)	$[t, t+1]$ (3)	$[t, t+10]$ (4)	$[t, t+1]$ (5)	$[t, t+10]$ (6)
Own EA $[r + 8]$	0.707*** (0.112)	1.512*** (0.059)	0.769*** (0.103)	1.232*** (0.054)	0.391*** (0.150)	0.945*** (0.083)
Own EA $[r + 9]$	0.679*** (0.110)	1.566*** (0.058)	0.745*** (0.097)	1.329*** (0.053)	0.476*** (0.160)	1.023*** (0.084)
Own EA $[r + 10]$	0.783*** (0.106)	1.576*** (0.058)	0.643*** (0.094)	1.402*** (0.053)	0.451*** (0.153)	1.145*** (0.082)
Own 8-K $[r]$	0.389*** (0.051)	0.044** (0.019)	0.228*** (0.045)	-0.006 (0.019)	0.376*** (0.075)	0.030 (0.030)
Own key development $[r]$	0.465*** (0.034)	0.133*** (0.016)	0.315*** (0.032)	0.091*** (0.015)	0.196*** (0.047)	0.080*** (0.023)
Positive news coverage $[r]$	0.340*** (0.049)	0.108*** (0.024)	0.241*** (0.046)	0.091*** (0.022)	-0.073 (0.075)	0.043 (0.035)
Negative news coverage $[r]$	-0.075 (0.049)	-0.061** (0.024)	-0.098** (0.046)	-0.057** (0.023)	0.001 (0.080)	0.011 (0.038)
log(1 + Increase in news volume)	0.398*** (0.044)	0.012 (0.019)	0.276*** (0.036)	-0.015 (0.017)	0.556*** (0.059)	0.028 (0.028)
log(1 + Decrease in news volume)	0.160*** (0.042)	0.034 (0.021)	0.125*** (0.038)	0.031 (0.020)	0.333*** (0.080)	0.041 (0.038)
Firm x Year-quarter FE	yes	yes	yes	yes	yes	yes
Observations	9,159,341	8,430,871	8,038,145	6,975,409	3,243,285	2,701,475
R-squared	.119	.398	.103	.376	.143	.494

This table shows how retail order imbalance varies in response to a memory cue. In columns 1 and 2, retail order imbalance from the Barber et al. (2024) (BHJOS) algorithm is regressed on the cue dummy and a battery of control variables. In columns 3 and 4, the dependent variable is retail order imbalance from the Boehmer et al. (2021) (BJZZ) algorithm. Columns 5 and 6 mirror columns 3 and 4, except that retail order imbalance is calculated using only odd lot trades (trades of fewer than 100 shares). All columns include the following control variables: 21 dummy variables that capture a potential own earnings announcement in the window $[r - 10, r + 10]$, a dummy variable that captures the filing of an own 8-K, a dummy variable that captures an own Key Development, a dummy variable each for positive/negative news coverage, and the logarithm of 1 plus the absolute increase or decrease in the volume of news coverage. All columns also include firm x year-quarter fixed effects. Standard errors are clustered by firm and trading day and are displayed in parentheses below the coefficients. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 3: Retail order imbalance results, part 1

(b) Table 3: Retail order imbalance results, part 2

6.5 Table 4: Cross-sectional cuts along firm size

Read:

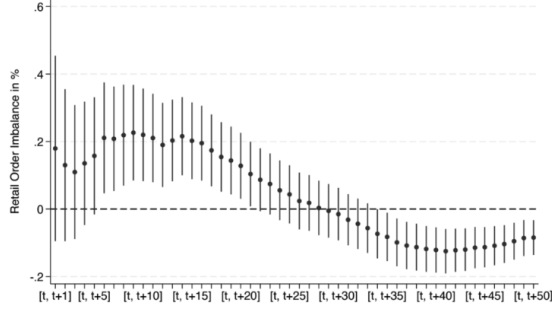


Figure 3

Retail order imbalance dynamics

This figure traces out the dynamics of retail order imbalance, constructed using the Barber et al. (2024) algorithm, in response to a memory cue by showing results from 50 separate regressions. The dependent variable in each regression is the average retail order imbalance over a different time window, ranging from $[t, t+1]$ to $[t, t+50]$. The dots in the figure represent the coefficients on the cue dummy, along with 95% confidence intervals, from each regression. All regressions include the full set of controls from Table 3 and firm-year-quarter fixed effects. Standard errors are clustered by firm and trading day.

Table 4

Cross-sectional cuts along firm size

Dependent variable:	Return $[t, t+10]$ (%)		
	All (1)	Small (2)	Large (3)
Cue (size of cueing firm > 90th ptile)	0.093*** (0.032)	0.168*** (0.062)	0.021 (0.032)
Cue (size of cueing firm in 51st - 90th ptile)	0.009 (0.024)	0.052 (0.048)	-0.036 (0.024)
Cue (size of cueing firm in 10th - 50th ptile)	-0.037 (0.023)	-0.057 (0.045)	-0.038* (0.023)
Cue (size of cueing firm < 10th ptile)	0.017 (0.035)	-0.054 (0.066)	0.068** (0.031)
Controls	yes	yes	yes
Firm x Year-quarter FE	yes	yes	yes
Observations	16,502,158	8,238,578	8,262,465
R-squared	.156	.161	.178

This table shows how the return effect of a memory cue varies with the size of the cueing and cued firm. In all columns, the dependent variable is the return over $[t, t+10]$. The main independent variables are four dummy variables that capture cues from firms at different points of the size distribution. Column 1 shows the results for the full sample, while columns 2 and 3 split the sample along the median market capitalization, and show the results for small and large cued firms, respectively. All columns include the following control variables: 21 dummy variables that capture a potential own earnings announcement in the window $[t-10, t+10]$, a dummy variable that captures the filing of an own S-K, a dummy variable that captures an own Key Development, a dummy variable each for positive/negative news coverage, and the logarithm of 1 plus the absolute increase or decrease in the volume of news coverage. All columns also include firm x year-quarter fixed effects. These fixed effects can result in singleton observations that are dropped during the estimation, which explains why the sum of the number of observations in columns 2 and 3 is slightly lower than the number of observations in column 1. Standard errors are clustered by firm and trading day and are displayed in parentheses below the coefficients. * $p < .1$; ** $p < .05$; *** $p < .01$.

- Table 4 splits cues by cueing-firm size and cued-firm size.
- Cues from firms above the 90th percentile of market capitalization produce a 0.093% return over $[t, t+10]$ in the full sample.
- For small cued firms, the effect of a cue from a top-10% cueing firm is 0.168%.
- For large cued firms, the same coefficient is only 0.021 and insignificant.
- Cues from smaller cueing firms are generally insignificant.

Detailed interpretation. This table matches two pieces of attention theory. First, large firms' earnings announcements are more visible, so they are better cues. Second, small stocks are more susceptible to retail flow and price pressure. A memory cue is therefore most powerful when a large salient cueing firm triggers recall of a small stock.

Economic message. The effect is strongest when attention is easiest to create and prices are easiest to move.

6.6 Table 5: Testing the psychological mechanism

Read:

- Panel A tests whether stronger attention during encoding strengthens the memory association.
- Strong associations based on joint high trading volume or joint retail trading volume produce larger return responses for large cueing firms.
- For high trading volume, the strong-cue coefficient is 0.176 while other large cues are 0.039; the equality test has a p-value around 0.023.
- Panel B tests similarity and interference: shared positive or negative surprises, extreme surprises, low interference, and market-hours versus after-hours cues.
- The strongest patterns support associative memory: more attention at encoding and more similar experiences strengthen recall.

Detailed interpretation. Table 5 is the most psychologically specific part of the paper. It goes beyond showing an average cue effect and asks whether the effect follows associative-memory comparative statics. Associations formed under high attention should be stronger. Associations formed by similar experiences

should also be stronger. Interference from other announcements should weaken recall. The evidence lines up with these predictions more than with a simple information-spillover account.

Economic message. The market effect behaves like a memory effect, not just a general earnings-announcement spillover.

Table 5
Testing the psychological mechanism

Panel A

Dependent variable:	Return $[t, t+10]$ in %				
	High trading volume (1)	High BHOJ retail volume (2)	High BJZZ retail volume (3)	High media coverage (4)	Top mover (5)
Association stronger due to joint:					
<i>Size of cueing firm > 90th pctile</i>					
Cue (strong association)	0.176*** (0.053)	0.248*** (0.078)	0.149* (0.088)	0.074* (0.039)	0.257 (0.566)
Cue (all other associations)	0.039 (0.035)	0.058* (0.033)	0.080** (0.033)	0.097** (0.046)	0.085*** (0.031)
<i>Size of cueing firm in 51st - 90th pctile</i>					
Cue (strong association)	0.031 (0.040)	0.047 (0.053)	-0.008 (0.066)	-0.005 (0.035)	-0.028 (0.210)
Cue (all other associations)	-0.020 (0.030)	-0.011 (0.027)	-0.001 (0.026)	0.001 (0.032)	-0.001 (0.025)
Controls	yes	yes	yes	yes	yes
Firm x Year-quarter FE	yes	yes	yes	yes	yes
Observations	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158
R-squared	.156	.156	.156	.156	.156
p-value (strong = other, large cues)	.0232	.0217	.4551	.6953	.7620
p-value (strong = other, small cues)	.2664	.2937	.9103	.8863	.8989

(Continued)

Table 5

Continued

Panel B

Dependent variable:	Return $[t, t+10]$ in %					
	Pos. surprise (6)	Neg. surprise (7)	Extreme pos. surprise (8)	Extreme neg. surprise (9)	Low interference (10)	EA market hours (11)
Association stronger due to joint:						
<i>Size of cueing firm > 90th pctile</i>						
Cue (strong association)	0.082 (0.057)	0.220 (0.143)	0.860 (1.994)	-5.055*** (0.645)	0.002** (0.041)	0.079* (0.045)
Cue (all other associations)	0.067** (0.035)	0.070** (0.031)	0.085*** (0.031)	0.066*** (0.031)	0.083* (0.048)	0.084** (0.034)
<i>Size of cueing firm in 51st - 90th pctile</i>						
Cue (strong association)	-0.012 (0.045)	0.385*** (0.093)	-0.268 (0.806)	-1.280 (0.874)	0.022 (0.032)	0.018 (0.037)
Cue (all other associations)	0.001 (0.027)	-0.016 (0.026)	-0.001 (0.025)	-0.001 (0.025)	-0.028 (0.040)	-0.012 (0.030)
Controls	yes	yes	yes	yes	yes	yes
Firm x Year-quarter FE	yes	yes	yes	yes	yes	yes
Observations	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158
R-squared	.156	.156	.156	.156	.156	.156
p-value (strong = other, large cues)	.9355	.2939	.8973	.0000	.8925	.9113
p-value (strong = other, small cues)	.7877	.0000	.7406	.1432	.3268	.9637

This table shows how the return effect of a memory cue varies with the strength of the underlying memory association. In all columns, the dependent variable is the return over $[t, t+10]$. The main independent variables are four dummies. The first dummy is equal to one if the cue targets a strong underlying memory association and if the cueing firm's size is above the 90th percentile. The column header indicates which features during the encoding determine whether the underlying association is strong. The second dummy captures all other cues from firms with size above the 90th percentile. The third and fourth dummies mirror the first and second, except that they capture the incremental effects of (strong) cues from firms in the 51st to the 90th percentile of size. All columns include the following control variables: 21 dummy variables that capture a potential own earnings announcement in the window $[t-10, t+10]$, a dummy variable that captures the filing of an own 8-K, a dummy variable that captures an own Key Development, two dummy variables that capture positive and negative news coverage, respectively, and the logarithm of 1 plus the absolute increase or decrease in the volume of news coverage. All columns also include firm x year-quarter fixed effects. Panel B is a continuation of panel A. Standard errors are clustered by firm and trading day and are displayed in parentheses below the coefficients. p -values for tests on the equality of the coefficients on first and second dummy, and the third and fourth dummies, are displayed at the bottom of the table. $^*p < .1$, $^{**}p < .05$, $^{***}p < .01$.

(a) Table 5, Panel A: attention during encoding

(b) Table 5, Panel B: similarity, interference, and timing

6.7 Table 6: Alternative cues

Read:

- Table 6 asks whether non-earnings events can act as cues.
- The key focus is events that plausibly create external attention, such as large earnings announcements, top-mover style events, or other salient information releases.
- The baseline memory result is strongest when the cue itself is an attention-grabbing event.
- Weaker or less salient cues have smaller effects, consistent with the idea that a cue must attract enough attention to trigger recall.

Detailed interpretation. Table 6 broadens the interpretation of the cue mechanism. It suggests that the earnings announcement is not special because of earnings content alone. It is useful because it is an attention event. Other salient cues can also trigger associated memories, but the effect depends on cue salience.

Economic message. Memory retrieval requires a salient trigger. Attention is the activation input for recall.

Table 6 Alternative cues Panel A						
Dependent variable:	Returns $[t, t+10]$ in %					
	als (1)	High trading volume (2)	High BUDCH small volume (3)	High BIZZ small volume (4)	High media coverage (5)	Top sector (6)
<i>Earnings announcement</i>						
Cue (baseline)	0.086** (0.034)					
Cue (strong association)	0.198*** (0.056)	0.294*** (0.063)	0.294*** (0.063)	0.148 (0.083)	0.034* (0.043)	0.294 (0.279)
Cue (all other associations)	0.033 (0.039)	0.027 (0.056)	0.087** (0.036)	0.087** (0.036)	0.087** (0.036)	0.087** (0.034)
<i>Key Development (accounting)</i>						
Cue (baseline)	0.019 (0.015)					
Cue (strong association)	0.029 (0.027)	0.045 (0.040)	-0.017 (0.060)	0.023 (0.021)	0.080** (0.043)	0.054 (0.246)
Cue (all other associations)	0.015 (0.018)	0.016 (0.046)	0.021 (0.016)	0.018 (0.025)	0.018 (0.015)	0.013 (0.016)
<i>Key Development (nonaccounting)</i>						
Cue (baseline)	0.014 (0.012)					
Cue (strong association)	0.038** (0.018)	0.057** (0.026)	0.026 (0.030)	0.020 (0.014)	0.095 (0.044)	0.095 (0.184)
Cue (all other associations)	0.004 (0.013)	0.009 (0.012)	0.013 (0.012)	0.008 (0.017)	0.014 (0.012)	0.014 (0.012)
<i>Firm R & K</i>						
Cue (baseline)	-0.000 (0.013)					
Cue (strong association)	0.003 (0.023)	0.006 (0.035)	0.000 (0.040)	-0.030* (0.017)	0.129 (0.138)	0.129 (0.138)
Cue (all other associations)	-0.002 (0.014)	-0.001 (0.013)	0.000 (0.013)	0.001 (0.013)	-0.001 (0.013)	-0.001 (0.013)
Controls	yes	yes	yes	yes	yes	yes
Firm x Year-quarter FE	yes	yes	yes	yes	yes	yes
Observations	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158
R-squared	.156	.156	.156	.156	.156	.156
p-value (strong = other, EA)	-	.0005	.0004	.4873	.6081	.7186
p-value (strong = other, KeyDev acc.)	-	.4568	.5110	.4426	.7040	.0517
p-value (strong = other, KeyDev nonacc.)	-	.0742	.8996	.6695	.5853	.6592
p-value (strong = other, R & K)	-	.8636	.8450	.9930	.0135	.3024

(a) Table 6: Alternative cues, part 1

Table 6 Continued Panel B								
Dependent variable:	Returns $[t, t+10]$ in %							
Association stronger due to joint:	Pos. surprise (1)	Neg. surprise (2)	Extreme pos. surprise (3)	Extreme neg. surprise (4)	Low interference (5)	EA market hours (12)	EA after hours (13)	EA other hours (14)
<i>Earnings announcement</i>								
Cue (strong association)	0.077 (0.061)	0.411*** (0.150)	0.779 (1.583)	-2.231*** (1.472)	0.108** (0.043)	0.088* (0.040)	0.095 (0.040)	0.095 (0.040)
Cue (all other associations)	0.087** (0.039)	0.074** (0.034)	0.087** (0.034)	0.087** (0.034)	0.087 (0.054)	0.084* (0.044)	0.084** (0.044)	0.084** (0.037)
<i>Key Development (accounting)</i>								
Cue (strong association)	0.011 (0.027)	0.012 (0.041)	-0.187 (2.050)	-1.164 (2.565)	0.020 (0.020)	0.014 (0.023)	0.054 (0.246)	0.054 (0.246)
Cue (all other associations)	0.021 (0.017)	0.019 (0.016)	0.019 (0.015)	0.019 (0.015)	0.003 (0.025)	0.021 (0.020)	0.013 (0.016)	0.013 (0.016)
<i>Key Development (nonaccounting)</i>								
Cue (strong association)	0.015 (0.018)	0.010 (0.046)	0.260 (0.611)	0.233 (0.871)	0.023 (0.014)	0.096 (0.015)	0.042* (0.023)	0.042* (0.023)
Cue (all other associations)	0.014 (0.013)	0.014 (0.012)	0.014 (0.012)	0.014 (0.012)	-0.000 (0.020)	0.020 (0.014)	0.008 (0.012)	0.008 (0.012)
<i>Firm R & K</i>								
Cue (strong association)	0.026 (0.022)	0.031 (0.057)	-2.054** (0.875)	-0.210 (1.181)	0.014 (0.017)	-0.015 (0.015)	-0.008 (0.027)	-0.008 (0.027)
Cue (all other associations)	-0.007 (0.015)	-0.001 (0.013)	0.000 (0.013)	-0.000 (0.013)	-0.016 (0.019)	0.000 (0.016)	0.001 (0.016)	0.001 (0.016)
Controls	yes	yes	yes	yes	yes	yes	yes	yes
Firm x Year-quarter FE	yes	yes	yes	yes	yes	yes	yes	yes
Observations	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158	16,502,158
R-squared	.156	.156	.156	.156	.156	.156	.156	.156
p-value (strong = other, EA)	.8766	.0213	.7264	.0003	.2580	.8672	.8607	.8607
p-value (strong = other, KeyDev acc.)	.7040	.2344	.6200	.6607	.2991	.7912	.2813	.2813
p-value (strong = other, KeyDev nonacc.)	.0770	.5225	.6620	.8015	.3164	.6605	.1148	.1148
p-value (strong = other, R & K)	.1911	.7383	.0152	.8521	.2203	.2861	.7789	.7789

This table shows how the main effect of a memory cue varies with the type of trading event as well as with the strength of the underlying memory association. In all columns, the dependent variable is the return over $[t, t+10]$. The independent variables in all columns capture trading events from firms above the 90th percentile of size. In column 1, the main independent variables are four dummies that capture the effects of different types of trading events: Earnings announcements, Key Development (accounting), Key Development (nonaccounting), and Filings of Form 8-K. In all other columns, the main independent variables are eight dummies, two for each type of trading event, that break out whether the trading event targets a strong underlying memory association. The column header indicates which features during the trading event determine whether the underlying association is strong. All columns include the following control variables: 21 dummy variables that capture a potential news coverage announcement in the window $[t-15, t+10]$, a dummy variable that captures the filing of an 8-K, a dummy variable that captures an own Key Development, two dummy variables that capture positive and negative news coverage, respectively, and the logarithm of 1 plus the absolute increase or decrease in the volume of news coverage. All columns also include Firm x year-quarter fixed effects. Panel B is a continuation of panel A. Standard errors are clustered by firm and trading day and are displayed in parentheses below the coefficients. For each type of trading event, the p-value from a test on the equality of the coefficients on the two dummies is displayed at the bottom of the table. * $p < .1$, ** $p < .05$, *** $p < .01$.

(b) Table 6: Alternative cues, part 2

6.8 Table 7: Anticipation of an earnings announcement + Table 8: Surprise of the cue

Read:

- Table 7 tests whether anticipation of an earnings announcement is enough to trigger the cue effect before the actual announcement.
- The pre-event coefficients for $t - 3$, $t - 2$, and $t - 1$ are small or insignificant, so anticipation alone does not explain the result.
- After-hours announcements do not produce the same effect on day t , supporting the role of real-time attention.
- Table 8 tests whether the cueing firm's earnings surprise predicts the cued firm's return response.
- Surprise magnitude does not map cleanly into the cued firm's return response, weakening a fundamental spillover explanation.

Detailed interpretation. Tables 7 and 8 jointly rule out two natural alternatives. First, if the effect came from predictable announcement timing rather than attention, returns should appear before the cue. They do not. Second, if the effect came from fundamental news spillovers, the cueing firm's surprise should predict the cued firm's return. The surprise tests do not support that simple information channel.

Economic message. The cue must be experienced as an attention event, and its informational content is not the main driver of the cued firm's return.

Table 7
Anticipation of earnings announcements

Return window:	$[t-3]$ (1)	$[t-2]$ (2)	$[t-1]$ (3)	$[t-1]$ (4)
Cue $[t]$	-0.012 (0.009)	-0.004 (0.009)	-0.003 (0.010)	
Cue after hours $[t-1]$				-0.005 (0.015)
Cue market hours $[t]$				-0.008 (0.011)
Controls	yes	yes	yes	yes
Firm x Year-quarter FE	yes	yes	yes	yes
Observations	16,501,071	16,501,152	16,501,645	16,501,645
R-squared	.015	.014	.014	.014

This table tests whether anticipation of a cueing event can lead to return effects. In columns 1 - 3, the return on days $t-3$, $t-2$, and $t-1$, respectively, is regressed on a dummy variable that captures a cue on day t . In column 4, the return on day $t-1$ is regressed on a dummy that captures a cue that occurred after the market closes on day $t-1$ as well as a dummy that captures a cue that occurred during market hours on day t . All columns include the following control variables: 21 dummy variables that capture a potential own earnings announcement in the window $[t-10, t+10]$, a dummy variable that captures the filing of an own 8-K, a dummy variable that captures an own Key Development, a dummy variable each for positive/negative news coverage, and the logarithm of 1 plus the absolute increase or decrease in the volume of news coverage. All columns also include firm x year-quarter fixed effects. Standard errors are clustered by firm and trading day and are displayed in parentheses below the coefficients. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 7: Anticipation of an earnings announcement

Table 8
Surprise of the cue

Dependent variable: Surprise:	Return $[t, t+10]$ in %			
	EW		VW	
	(1)	(2)	(3)	(4)
Cue $[t]$	0.089** (0.035)		0.090** (0.035)	
Cue x Surprise $[t]$	-7.678 (18.797)		-10.606 (18.246)	
Cue x Surprise quintile 1 $[t]$		0.029 (0.047)		0.022 (0.047)
Cue x Surprise quintile 2 $[t]$		-0.030 (0.142)		0.124 (0.141)
Cue x Surprise quintile 3 $[t]$		0.175*** (0.062)		0.169*** (0.061)
Cue x Surprise quintile 4 $[t]$		0.172*** (0.060)		0.151*** (0.060)
Cue x Surprise quintile 5 $[t]$		0.028 (0.067)		0.046 (0.066)
Controls	yes	yes	yes	yes
Firm x Year-quarter FE	yes	yes	yes	yes
Observations	16,502,158	16,502,158	16,502,158	16,502,158
R-squared	.156	.156	.156	.156

This table tests whether the earnings surprise of the cueing firm(s) can predict the return response of the cued firm. The earnings surprise is the difference between the actual earnings announced by the cueing firm and the median analyst earnings forecast, scaled by the share price of the firm from three trading days prior to the announcement. If there are multiple cues for firm i on day t , columns 1 and 2 use the equally weighted average surprise, and columns 3 and 4 use the value-weighted average surprise. In columns 1 and 3, the cue dummy is interacted with the surprise, while in columns 2 and 4, the cue dummy is interacted with dummy variables for each quintile of the surprise distribution. All columns include the following control variables: 21 dummy variables that capture a potential own earnings announcement in the window $[t-10, t+10]$, a dummy variable that captures the filing of an own 8-K, a dummy variable that captures an own Key Development, a dummy variable each for positive/negative news coverage, and the logarithm of 1 plus the absolute increase or decrease in the volume of news coverage. All columns also include firm x year-quarter fixed effects. Standard errors are clustered by firm and trading day and are displayed in parentheses below the coefficients. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 8: Surprise of the cue

6.9 Table 9: Trading strategy over time + Table 10: Trading strategy: Long and short leg

Read:

- Table 9 forms a long-short strategy using small firms: long firms with a cueing event and short firms without one.
- The strategy is not profitable in 2005–2009, has positive but insignificant alpha in 2010–2014, and earns 0.693% over $[t, t+10]$ in 2015–2019.
- The 2015–2019 alpha is significant at the 1% level and corresponds to a large annualized abnormal return.
- Table 10 decomposes the 2015–2019 strategy into long and short legs.
- The long leg earns 0.467% over $[t, t+1]$ and 0.853% over $[t, t+10]$, while the short leg has insignificant alpha.

Detailed interpretation. These tables are not presented as a low-cost implementable arbitrage, because shorting many small stocks could be costly. They are asset-pricing robustness tests. The factor-adjusted alpha shows that the cue effect is not easily explained by standard risk factors. The long-leg decomposition is especially important: the memory mechanism predicts buying pressure in cued stocks, so the alpha should come from the long side. Table 10 shows that it does.

Economic message. The predictable return is a buying-pressure phenomenon, not a standard factor exposure or short-leg artifact.

Table 9**Trading strategy over time**

Return window:	$[t, t+10]$		
Time period:	2005 - 2009 (1)	2010 - 2014 (2)	2015 - 2019 (3)
Alpha [%]	-0.265 (0.414)	0.224 (0.219)	0.693*** (0.228)
Mkt	0.292 (0.329)	-0.027 (0.264)	0.124 (0.287)
SMB	0.087 (0.569)	-0.163 (0.477)	0.637 (0.456)
HML	-0.539 (0.599)	-0.141 (0.529)	0.573 (0.486)
Momentum	-0.144 (0.443)	0.233 (0.363)	0.162 (0.341)
ST reversal	0.249 (0.340)	0.211 (0.410)	0.487 (0.412)
Observations	374	516	507
R-squared	.007	.002	.010

This table shows how the trading strategy performs in three different sets of 5 years each. The trading strategy return in each time period is regressed on the market, size, value, momentum, and short-term reversal factors, which are sourced from the Kenneth French Data Library. The trading strategy return is the return of a long-short strategy. The long leg of the strategy consists of a value-weighted portfolio of stocks with a cueing event on day t , and the short leg of the strategy consists of a value-weighted portfolio of stocks without a cueing event on day t . The weights in these portfolios are the market capitalization of each stock on day $t-3$. These portfolios are formed using firms whose market capitalization from $t-3$ is below the median. Standard errors are displayed in parentheses below the coefficients. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 9: Trading strategy over time

Table 10**Trading strategy: Long and short leg**

Return window:	$[t, t+1]$	$[t, t+10]$	$[t, t+1]$	$[t, t+10]$
Strategy:	Long leg (1)	(2)	Short leg (3)	(4)
Alpha [%]	0.467** (0.184)	0.853*** (0.271)	-0.028 (0.046)	0.160 (0.147)
Mkt	0.776*** (0.232)	0.609* (0.342)	0.908*** (0.058)	0.485*** (0.186)
SMB	1.361*** (0.369)	1.814*** (0.543)	0.719*** (0.092)	1.177*** (0.295)
HML	0.302 (0.393)	0.406 (0.578)	0.000 (0.098)	-0.167 (0.315)
Momentum	-0.192 (0.276)	-0.463 (0.406)	-0.252*** (0.069)	-0.626*** (0.221)
ST reversal	0.540 (0.333)	0.742 (0.490)	0.174** (0.083)	0.254 (0.267)
Observations	507	507	507	507
R-squared	.075	.053	.472	.089

This table shows how the long and short leg of the trading strategy perform in 2015–2019. Returns of the long and short leg are regressed on the market, size, value, momentum, and short-term reversal factors, which are sourced from the Kenneth French Data Library. In columns 1 and 2, the dependent variable is the return of the long leg of the strategy, which consists of a value-weighted portfolio of stocks with a cueing event on day t . In columns 3 and 4, the dependent variable is the return of the short leg of the strategy, which consists of a value-weighted portfolio of stocks without a cueing event on day t . The weights in these portfolios are the market capitalization of each stock on day $t-3$. These portfolios are formed using firms whose market capitalization from $t-3$ is below the median. Standard errors are displayed in parentheses below the coefficients. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 10: Trading strategy - long and short leg

7 Short summary

- The paper shows that internally generated attention from memory associations can move stock prices.
- Calendar rotations of Pattern-firm earnings schedules create quasi-random overlapping announcement contexts.
- Cues produce positive abnormal returns for associated firms over short horizons.
- The price effect builds over one to two weeks and then reverses, consistent with temporary attention-driven price pressure.
- Retail order imbalance rises after cues and follows the same timing as returns.
- Large cueing firms and small cued firms generate the strongest effects.
- Stronger encoding attention, greater similarity, lower interference, and more salient cue events strengthen the effect.
- Anticipation and earnings-surprise tests weaken fundamental-information spillover explanations.
- A cue-based strategy earns significant alpha in 2015–2019, driven by the long leg.
- The paper provides market-level evidence that cognitive memory mechanisms affect asset prices.

8 Conclusion

8.1 Core conclusion

Charles shows that memory-induced attention can distort financial prices. When a salient event at one firm cues investors to recall an associated firm, retail buying pressure can spill over to the associated firm's stock and generate temporary abnormal returns.

8.2 Economic intuition

Associative memory links items that are experienced in similar contexts. When one item is later cued, associated items are more likely to be recalled. In this paper, earnings-announcement overlaps create the memory association, and a later earnings announcement triggers recall. Since retail investors tend to buy attention-grabbing stocks, recall generates buying pressure.

8.3 Identification contribution

The paper’s main identification contribution is the use of rigid earnings schedules and calendar rotations. This creates memory associations that are plausibly generated by calendar mechanics rather than by fundamental relatedness or discretionary announcement timing. The exclusions for industries, vertical links, and analyst overlap further sharpen the design.

8.4 Interpretation of the evidence

The evidence is strongest because several independent patterns point to the same mechanism. Returns rise and reverse. Retail buying rises and later fades. The effect is stronger when memory theory predicts stronger associations. The effect is concentrated where retail price pressure should matter most. Anticipation and surprise tests weaken alternative information explanations.

8.5 Asset-pricing implications

The paper expands the attention literature by showing that not all attention sources are external. Investors’ internal memory networks can shape return predictability and transient mispricing. This provides a microfoundation for some categorization and comovement effects in financial markets.

8.6 Limitations

The paper infers memory from market-level data rather than directly observing investors’ mental associations. The design focuses on earnings-announcement contexts and Pattern firms, so other types of memory associations may behave differently. Trading-strategy returns may not be implementable after shorting costs and other frictions.

8.7 Takeaway

Markets can be moved by what investors remember, not only by what investors see. Memory is an internal attention technology, and when many investors share similar contexts, memory can aggregate into measurable price pressure.